

Breast Cancer Diagnosis Using Machine Learning and Visualization-Based Feature Engineering

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1. Abstract

Breast cancer diagnosis is an important classification problem where accurate prediction can support early decision-making. This project applies machine learning and visualization-based analysis to the Breast Cancer Wisconsin Diagnostic dataset, which contains measurements from breast mass samples classified as malignant or benign.

The dataset was split into training, validation, and test sets using a 70/15/15 ratio to reduce data leakage. Logistic Regression was used as the baseline model, while SVM and Random Forest were trained for comparison. All models achieved strong results, but Random Forest slightly outperformed the others, with an accuracy of 0.9882, precision of 0.9815, recall of 1.0, F1-score of 0.9907, and ROC-AUC of 1.0. Therefore, Random Forest was selected for detailed evaluation. Visualizations such as the confusion matrix, ROC curve, precision-recall curve, feature importance plot, learning curve, PCA plot, and error analysis showed that the model provides strong and interpretable performance for breast cancer classification.

2. Introduction

Breast cancer is one of the most significant health problems worldwide, and early diagnosis plays an important role in improving treatment outcomes. In medical decision-making, classification models can help analyze diagnostic measurements and predict whether a tumor is malignant or benign. Machine learning provides useful methods for identifying patterns in medical data and supporting diagnostic prediction.

This project focuses on applying machine learning models to classify breast cancer cases using the Breast Cancer Wisconsin Diagnostic dataset. The dataset is suitable for this project because it is a real healthcare dataset, contains numerical diagnostic features, and supports binary classification. The dataset contains features computed from digitized images of fine needle aspirate samples of breast masses, describing characteristics of cell nuclei present in the image.

The project also emphasizes the role of visualization in the analytical process. According to the project requirements, students must use visualization to explore distributions, detect missing values and outliers, identify correlations, motivate feature engineering, and evaluate model performance using visual tools such as confusion matrices, ROC curves, precision-recall curves, feature importance plots, learning curves, clustering plots, and error analysis visualizations.

The main objectives of this project are:

1. To build machine learning models for classifying breast cancer cases as malignant or benign.
2. To use visualization to understand the dataset and support feature engineering.

3. To compare Logistic Regression, Support Vector Machine, and Random Forest models.
4. To select the best model using validation performance.
5. To evaluate and interpret the best model using visualization-based analysis.

3. Related Work

Machine learning has been widely used in medical diagnosis because it can learn patterns from clinical and diagnostic features. In breast cancer classification, machine learning models can help identify whether a tumor is malignant or benign based on numerical measurements extracted from medical images. The Breast Cancer Wisconsin Diagnostic dataset is a well-known benchmark dataset for this classification problem. It contains features computed from digitized images of breast mass samples and is commonly used for binary classification between malignant and benign tumors [1].

Logistic Regression is commonly used as a baseline model for binary classification tasks. It is simple, interpretable, and useful for estimating the probability that an observation belongs to a specific class [2]. In this project, Logistic Regression was used as the baseline model to provide a benchmark for comparing more advanced models.

Support Vector Machine is another widely used classification algorithm. Cortes and Vapnik introduced support-vector networks as a learning method for two-group classification problems [3]. The method attempts to find a decision boundary that separates classes effectively, making it useful for classification problems with complex feature relationships.

Random Forest is an ensemble learning method introduced by Breiman [4]. It combines multiple decision trees, where each tree is built using randomization, and the final classification is based on the combined output of the trees. Random Forest is useful because it can handle nonlinear relationships, reduce overfitting compared with a single decision tree, and provide feature importance values for interpretation [4].

Visualization-based model evaluation is also important in classification tasks. Confusion matrices show the number of correct and incorrect predictions for each class. ROC curves evaluate the relationship between the true positive rate and false positive rate across different thresholds [5], while precision-recall curves help analyze the balance between precision and recall [6]. In this project, these visualizations were used to interpret the performance of the selected Random Forest model.

4. Dataset Description

The dataset used in this project is the Breast Cancer Wisconsin Diagnostic dataset. It is a healthcare classification dataset from the UCI Machine Learning Repository. The dataset

contains diagnostic features computed from digitized images of fine needle aspirate samples of breast masses. These features describe characteristics of cell nuclei in the image.

Item	Description
Dataset name	Breast Cancer Wisconsin Diagnostic
Domain	Healthcare / medical diagnosis
Number of instances	569
Number of features	30 numerical features
Target variable	Diagnosis
Task type	Binary classification
Classes	Malignant and benign

The feature columns in the dataset are:

Feature Group	Example Features
Mean measurements	radius1, texture1, perimeter1, area1, smoothness1
Standard error measurements	radius2, texture2, perimeter2, area2, smoothness2
Worst/largest measurements	radius3, texture3, perimeter3, area3, smoothness3

5. Methodology

5.1 Data Preparation

The dataset was first loaded into pandas DataFrames. The feature matrix was stored as x , and the target variable was stored as y . The feature and target data were combined into a single DataFrame for exploration and preprocessing.

The diagnosis column was encoded into numerical values:

- $M = \text{Malignant} = 0$
- $B = \text{Benign} = 1$

This encoding allowed the diagnosis variable to be used in machine learning models.

5.2 Train, Validation, and Test Split

To avoid data leakage, the dataset was split before model training, scaling, and final model evaluation. A 70/15/15 split was used:

Set	Purpose	Percentage
Training set	Train the models	70%
Validation set	Tune and compare models	15%
Test set	Final evaluation only	15%

The test set was kept separate and was not used for training, feature selection, or model tuning. This is important because using the test set during model development can lead to overly optimistic results.

5.3 Visualization-Based Feature Engineering

Visualization was used as part of the analytical process, not only for final presentation. This follows the project requirement that students should use visualization to support feature engineering and model interpretation.

The following visualization-based steps were used:

Analytical Goal	Visualization Used	Purpose
Explore distributions	Histograms	Understand feature spread and skewness
Detect missing values	Missing value heatmap	Check data completeness
Detect outliers	Boxplots	Identify extreme feature values
Identify relationships	Correlation heatmap	Study relationships between features
Support feature selection	Correlation and feature importance plots	Identify useful predictors
Support dimensionality reduction	PCA plot	Visualize class separation in two dimensions

Some ratio-based features were created to capture relationships between tumor measurements. Examples include:

- $\text{area_per_radius} = \text{area1} / \text{radius1}$
- $\text{perimeter_per_radius} = \text{perimeter1} / \text{radius1}$
- $\text{compactness_per_smoothness} = \text{compactness1} / \text{smoothness1}$
- $\text{concavity_per_compactness} = \text{concavity1} / \text{compactness1}$
- $\text{concave_points_per_concavity} = \text{concave_points1} / \text{concavity1}$

After creating the new ratio features, the original features used to create them were removed to reduce redundancy. These included: `area1`, `radius1`, `perimeter1`, `compactness1`, `smoothness1`, `concavity1`, and `concave_points1`.

5.4 Feature Scaling

Feature scaling was applied after feature engineering. The scaler was fitted only on the training set, then applied to the validation and test sets. This prevents information from the validation or test sets from influencing the training process.

Scaling was especially important for Logistic Regression and SVM because these models are sensitive to feature magnitude. Although Random Forest does not require scaling, the same processed feature set was used for consistency across models.

5.5 Machine Learning Models

Three models were trained and compared:

Model	Role
Logistic Regression	Baseline model
Support Vector Machine	Traditional machine learning model
Random Forest	Traditional ensemble model and final selected model

Logistic Regression was used as the baseline model. SVM and Random Forest were trained and tuned using the validation set. The best model was selected based on validation performance, especially F1-score and recall.

6. Results & Analysis

6.1 Machine Learning Models

The model comparison results are shown below:

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression Baseline	0.9765	1.0000	0.9623	0.9808	1.0000
SVM	0.9882	1.0000	0.9811	0.9905	1.0000
Random Forest	0.9882	0.9815	1.0000	0.9907	1.0000

The comparison shows that all three models achieved strong classification performance. Logistic Regression performed well as a baseline model, achieving an accuracy of 0.9765 and an F1-score of 0.9808. This indicates that even a simple linear model was able to classify the dataset effectively.

SVM and Random Forest both achieved the highest accuracy of 0.9882. However, Random Forest achieved a slightly higher F1-score of 0.9907, compared with 0.9905 for SVM. Random Forest also achieved the highest recall score of 1.0000. Because the difference between SVM and Random Forest was small, Random Forest was selected as the best model only by a slight

margin. It was chosen for final evaluation because it provided the best overall balance between recall and F1-score, while also supporting feature importance analysis.

6.2 Confusion Matrix Analysis

The confusion matrix for the Random Forest model (best model) shows the following results:

Actual Class	Predicted Malignant	Predicted Benign
Malignant	26	6
Benign	2	52

The confusion matrix shows that the Random Forest model correctly classified 26 malignant cases and 52 benign cases. It misclassified 6 malignant cases as benign and 2 benign cases as malignant.

This visualization is important because it shows the exact types of errors made by the model. In a medical diagnosis context, predicting a malignant tumor as benign is more serious than predicting a benign tumor as malignant, because it may delay treatment. Therefore, the 6 malignant cases predicted as benign represent the most important error type to consider.

Overall, the confusion matrix shows that the Random Forest model performed strongly, correctly classifying most cases. However, the presence of false benign predictions suggests that further threshold tuning or model improvement could be useful if the main goal is to reduce missed malignant cases.

6.3 ROC Curve and AUC Analysis

The ROC curve for the Random Forest model achieved: $AUC = 0.981$.

The ROC curve evaluates the model's ability to distinguish between malignant and benign cases across different classification thresholds. The curve is located close to the top-left corner of the plot, which indicates strong classification performance. The dashed diagonal line represents random guessing, while the Random Forest curve is far above this line.

An AUC value of 0.981 indicates that the model has excellent discriminatory ability. This means the Random Forest model can separate malignant and benign cases very effectively across different thresholds. The ROC curve confirms that the selected model performs much better than random classification.

6.4 Precision-Recall Curve Analysis

The precision-recall curve for the Random Forest model achieved: 0.990.

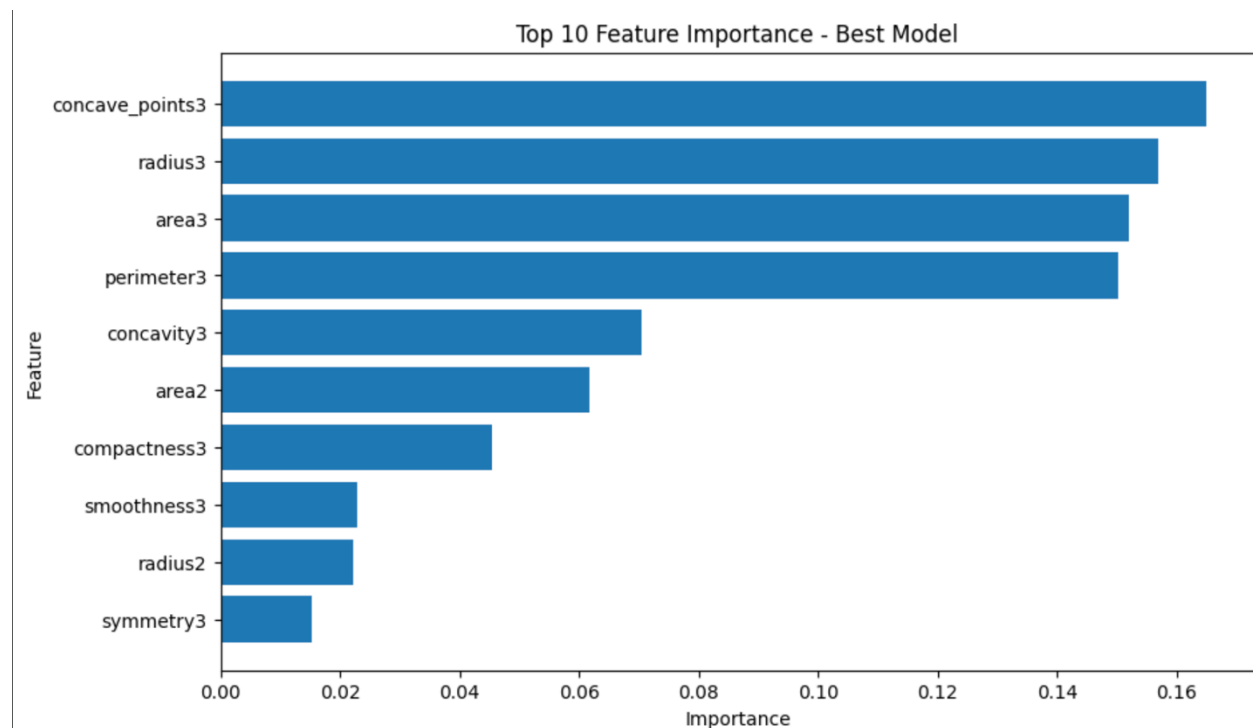
The precision-recall curve shows the relationship between precision and recall at different thresholds. The curve remains close to the upper area of the plot, which indicates strong

performance. A high average precision score of **0.990** means that the model maintains a strong balance between identifying positive cases and avoiding false predictions.

This visualization is useful because precision and recall are especially important in medical classification. High recall means the model is able to detect most cases of the target class, while high precision means that when the model predicts a class, it is usually correct. The result suggests that Random Forest provides reliable classification performance.

6.5 Feature Importance Analysis

The feature importance plot shows the top 10 most important features used by the Random Forest model. The most important features were:



The most influential feature was `concave_points3`, followed by `radius3`, `area3`, and `perimeter3`. These features belong mostly to the “worst” measurement group, represented by the suffix 3. This means that the largest or most severe tumor measurements were highly important in distinguishing malignant and benign cases.

The feature importance visualization helps interpret the model by showing which diagnostic measurements contributed most to prediction. It also supports feature selection because less important features can be reviewed or removed in future experiments to simplify the model.

6.6 Prediction Error Summary

The prediction error summary shows that the Random Forest model made far more correct predictions than incorrect predictions. According to the plot, the model produced approximately:

Prediction Type	Number of Samples
Correct	78
Incorrect	8

This visualization provides a simple summary of model performance. It confirms that most test samples were classified correctly. The incorrect predictions match the total number of errors shown in the confusion matrix:

- 6 false benign predictions + 2 false malignant predictions = 8 total errors

Since this is a classification task, a prediction error summary was used instead of a traditional residual plot. Residual plots are mainly used for regression problems, while classification error plots are more suitable for evaluating incorrect class predictions.

6.7 Learning Curve Analysis

The learning curve compares the training F1-score and validation F1-score as the training set size increases. The training score remained very high, close to 1.000, while the validation score improved as more training data was added.

At smaller training sizes, the validation score was lower, around 0.961, which suggests that the model benefited from more data. As the training size increased, the validation score improved to around 0.976. The gap between the training and validation curves indicates a small degree of overfitting, because the training score is consistently higher than the validation score.

However, the validation score remains high, which means the model generalizes well overall. The learning curve suggests that the Random Forest model performs strongly, but additional training data or further tuning may help reduce the gap between training and validation performance.

6.8 PCA Clustering Plot Analysis

The PCA clustering plot reduces the test data into two principal components for visualization. The plot shows benign and malignant samples in a two-dimensional space.

The PCA plot shows that many benign and malignant samples form partially separated groups. Benign cases are more concentrated in one region, while malignant cases are more spread out across the plot. Some overlap is visible between the two classes, which explains why the model still made a small number of classification errors.

This visualization is useful because it shows that the two classes have some natural separation in the feature space. However, because the separation is not perfect, machine learning models are needed to learn more complex boundaries using the full feature set.

6.9 Actual vs Predicted Class Distribution

The actual-versus-predicted class distribution plot shows how the Random Forest model classified each actual class.

For actual malignant cases, most were predicted as malignant, but some were predicted as benign. For actual benign cases, most were predicted as benign, with only a small number predicted as malignant.

This visualization supports the confusion matrix interpretation. It shows that the model performed very well for benign cases and also performed strongly for malignant cases, although some malignant samples were misclassified as benign. This type of visualization helps explain the model's error pattern in an easier way than metrics alone.

7. Conclusion & Future Work

This project applied machine learning and visualization-based analysis to the Breast Cancer Wisconsin Diagnostic dataset. The main goal was to classify breast cancer cases as malignant or benign using diagnostic features extracted from breast mass samples.

The dataset was prepared carefully by encoding the target variable, creating ratio-based features, removing redundant original features, scaling the final feature set, and splitting the data into training, validation, and test sets. The split was performed early in the workflow to reduce the risk of data leakage.

Three models were trained and compared: Logistic Regression, Support Vector Machine, and Random Forest. Logistic Regression performed well as a baseline model, while SVM and Random Forest achieved stronger performance. Random Forest was selected as the best model by a small margin because it achieved the highest F1-score and recall in the comparison table.

The visualization-based evaluation showed that the Random Forest model performed strongly. The ROC curve achieved an AUC of 0.981, and the precision-recall curve achieved an average precision of 0.990. The confusion matrix showed that the model correctly classified most cases, although it misclassified some malignant cases as benign. The feature importance plot showed that `concave_points3`, `radius3`, `area3`, and `perimeter3` were among the most important predictors. The learning curve showed strong generalization, while the PCA clustering plot showed partial separation between malignant and benign samples.

Overall, the project demonstrates that machine learning combined with visualization-based feature engineering and model evaluation can provide accurate and interpretable results for breast cancer classification. Future work could focus on threshold adjustment to reduce malignant cases misclassified as benign, testing additional models such as XGBoost, and using cross-validation to improve reliability.

8. References

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